Haotian Liu

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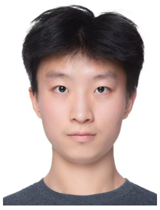

RESEARCH INTERESTS

LLM Agents | Multimodal Large Language Models | AI for Software Engineering and Human–Computer Interaction

EDUCATION

Xiamen University (XMU), Xiamen, China Sep. 2023 – Jun. 2027 (expected)
B.S. in Software Engineering | Overall Average: 82.50/100
Relevant Coursework: Machine Learning for Decision-Making and Optimization (96); Algorithms for Modern AI (94); Introduction to Artificial Intelligence (93); Interactive Design (90); Data Warehouse (90).

PUBLICATIONS

PilotBench: A Benchmark for General Aviation Agents with Safety Constraints.
IJCNN 2026, **accepted**. Co-first author. arXiv:2604.08987
Long-Horizon Memory Governance for LLM Agents.
Under review at EMNLP 2026. First author.
ProcuraClaw: Context-Aware Procurement Search via Adaptive Hotel Profiling and Centralization Rate Prediction.
Under review at CIKM 2026. Co-first author.

RESEARCH PROJECTS

DDRG: Reasoning-Graph Verification and Repair Apr. 2026 – Present
Research Collaborator | Li Lab
- To address majority-vote failures in sampled reasoning, introduced an inference-time graph verification method that aligns local claims, isolates answer-sensitive conflicts, and accepts only evidence-supported repairs.
- Evaluated LLaMA-3.1-70B on five reasoning benchmarks; case-level audits identified parser, verifier, and repair-selection failure modes and motivated conservative answer-selection gates.

ConcordCoder: Pre-generation Alignment for AI Coding Mar. 2026 – Present
Online Research Intern | Waseda Institute for Advanced Study
- To prevent intent and compatibility errors before AI-generated patches, proposed **ConcordCoder**, which reconstructs code-grounded task context and asks developers to confirm constraints, risks, and non-goals before generation.
- Designed an inspectable **AlignmentRecord** and a controlled evaluation of comprehension, review accuracy, evidence grounding, guard violations, and multi-round constraint retention.

NeuroMark: Multimodal Clinical Reasoning Sep. 2025 – Present
Research Contributor | Research Training at the School of Data Science, Fudan University
- Led the initial project phase and introduced a difficulty-discrimination analysis to compare multimodal clinical reasoning beyond raw accuracy; benchmarked 36 foundation models across 10+ datasets and professional examinations.
- Reproduced 3D diagnostic pipelines on IntrA and TOF-MRA data and developed weak-label strategies for scalable medical-image annotation.

Long-Horizon Memory Governance for LLM Agents (LHMG) Dec. 2025 – May 2026
Project Lead | Research Training at the School of Data Science, Fudan University
- To prevent silent overwrite drift in long-horizon agents, proposed an auditable memory-governance layer combining append-only revisions, active-state semantics, risk-gated conditioning, and governed forgetting.
- Increased GPT-4o cross-turn decision consistency by 13.3 points in controlled long-horizon evaluations.

PilotBench: Safety-Critical Aviation Agents Dec. 2025 – Mar. 2026
Project Lead | Research Training at the School of Data Science, Fudan University
- To evaluate LLM agents under physical and safety constraints, constructed PilotBench from 708 real trajectories, 34-channel telemetry, and nine flight phases, and evaluated 41 models.
- Introduced Pilot-Score and revealed a precision–controllability dichotomy plus a dynamic-complexity gap in high-workload phases; accepted by IJCNN 2026.

INDUSTRY EXPERIENCE

ProcuraClaw: Context-Aware Procurement Search Nov. 2025 – Apr. 2026
AI Research Intern | Huazhu Group
- To overcome vocabulary mismatch and hotel-specific ranking needs, built a deployed procurement-search agent with LLM entity extraction, hybrid retrieval, profile-conditioned scoring, and Thompson Sampling.
- Achieved MRR 0.95 and NDCG@5 0.91 on 175 evaluation queries; live A/B tests improved first-hit CTR by 6.2% and 9.8% across two deployment slices.

TECHNICAL SKILLS AND LANGUAGES

Programming and ML: Python, PyTorch, Transformers, SQL, Java; LLM agents, RAG, multimodal learning, evaluation, and information retrieval.
Research Tools: Git, Linux, Docker, LaTeX; experiment design, ablation studies, reproducible evaluation, and academic writing.
Languages and Tests: Chinese (native); English (TOEFL iBT 90); Japanese (working reading proficiency). GRE scheduled for Aug. 2026.